V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning

META

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Introduction

Motivation: Why World Models?

- Humans can:
  - Adapt to new tasks
  - Predict effects of actions
  - Perform goal-directed planning

- AI agents should similarly:
  - **Understand** physical world
  - **Predict** future states
  - **Plan** in unseen environments

Introduction

Previous approaches and limitations:

1. Classical predictive world models
    - Require: state + action + reward (interaction data)
    - Limitation: poor scalability
2. Video generation-based models
    - Trained on internet-scale video
    - Limitation:
        - Weak at robot control/planning
        - Expensive
        - Limited evaluation of planning (focusing on prediction/visual quality)

- Existing methods fail to scale to both **representation learning** **&** **control**

Introduction

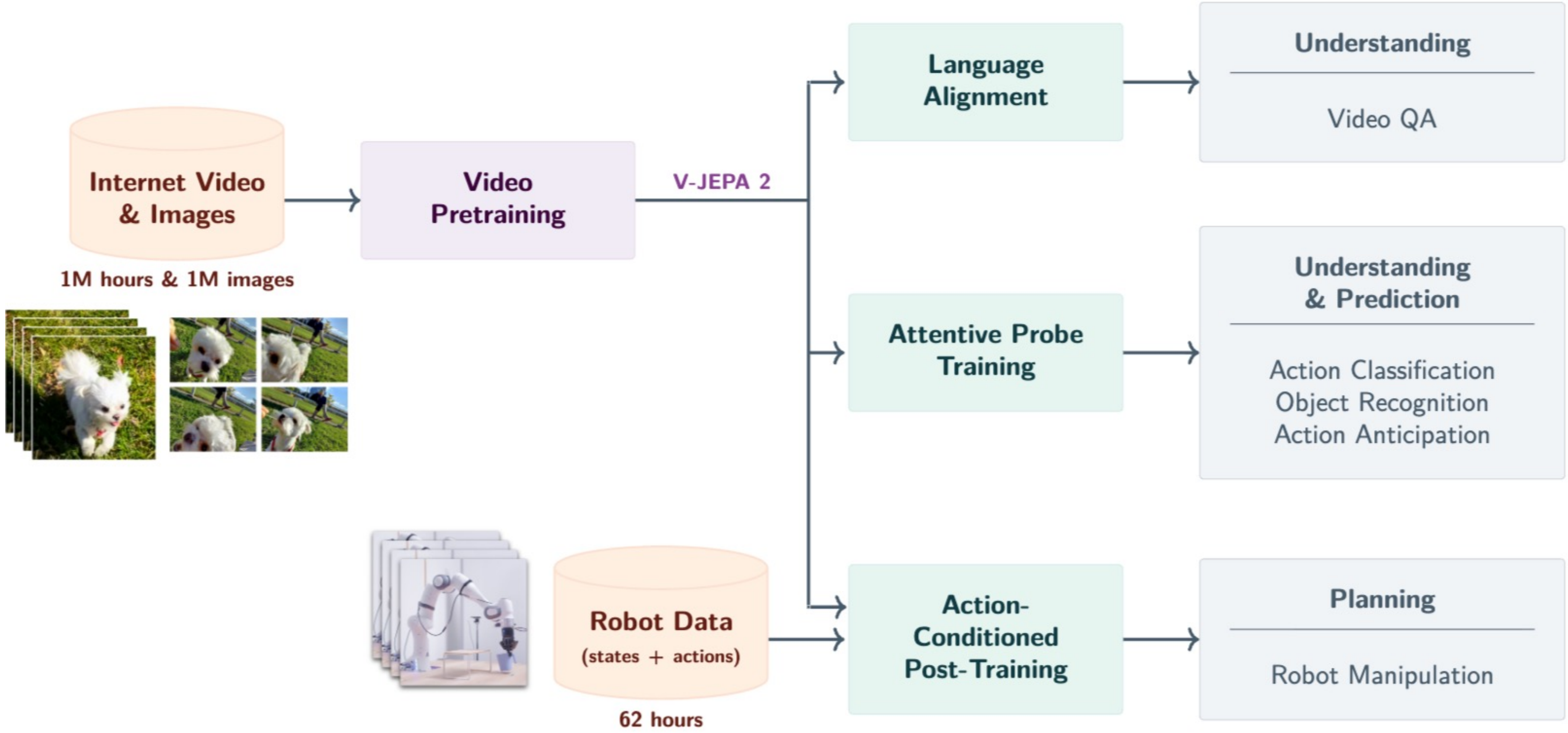
Key Idea: **Self-supervised World Model** from video

- Hypothesis: World knowledge can be learned from observation

Method Backbone: **JEPA (Joint-Embedding Predictive Architecture)**

- Predict in representation space (not pixels)
- Contrast:
  - Video generation → pixel-level prediction
  - JEPA → **semantic-level prediction**

V-JEPA 2 Overview



V-JEPA 2 Overview

1. Self-Supervised Pretraining (V-JEPA 2)
    - Input: internet-scale video (action-free)
    - Task: mask-denoising feature prediction
2. Action-Condition World Model (V-JEPA 2-AC)
    - Small interaction dataset
    - Learns:
        - Next state representation prediction
        - Conditioned on action + state
3. Planning
    - Uses learned world model

V-JEPA2: Self-Supervised Video Pretraining

Key Idea:

- Learn video representation from internet-scale video (1M hours)
- No labels, no actions → self-supervised learning

Training Objective:

- Masked video patches → **predict representation space**
- Not pixel reconstruction

V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

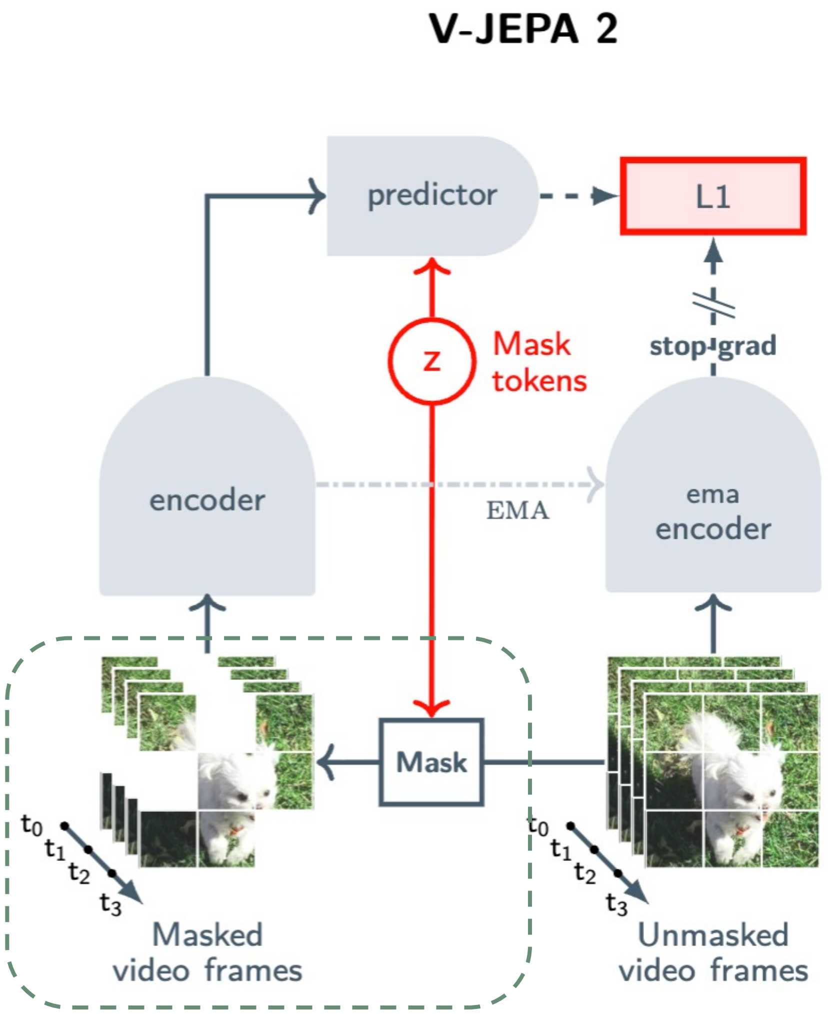
(1) Patchify
- Video clip ⟶ sequence of tokens (tubelets)

(2) Masking
- Randomly drop a subset of tokens

(3) Encoder
- Extract representations from visible patches

(4) Predictor
- Input: visible representations + mask tokens
- Output: predicted representation of masked patches

(5) EMA Encoder
- Extract target representations from the original video



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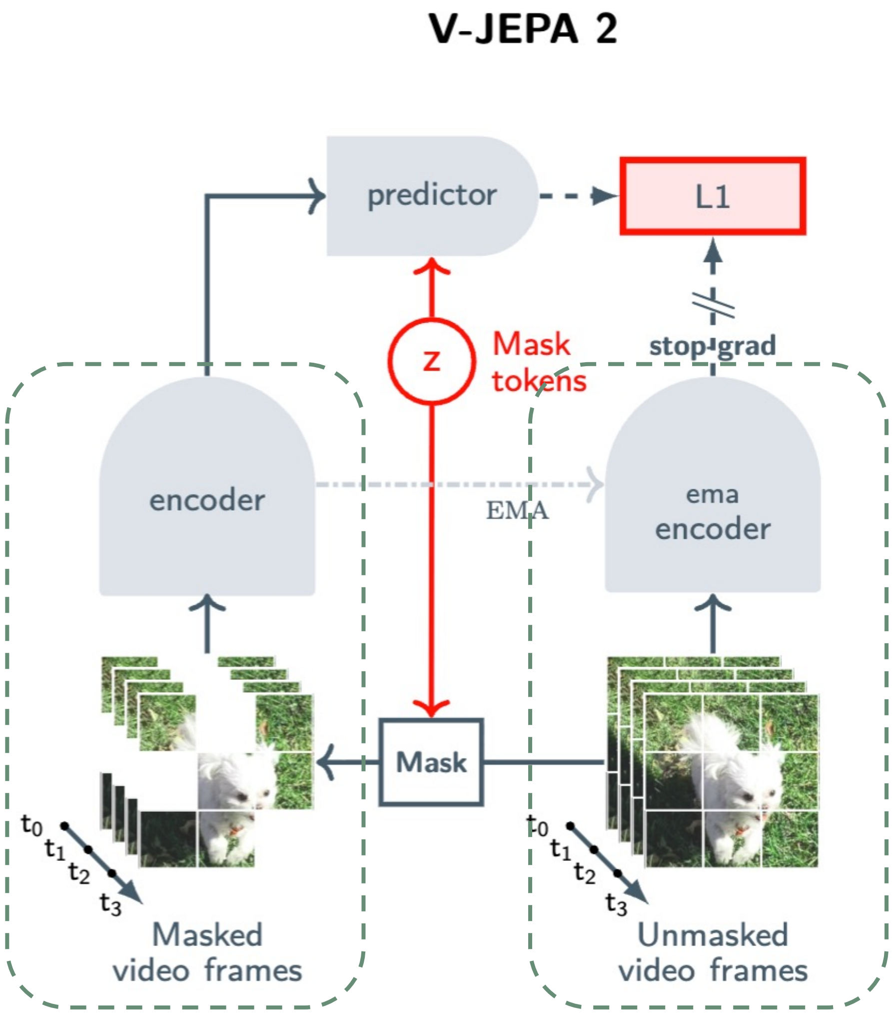

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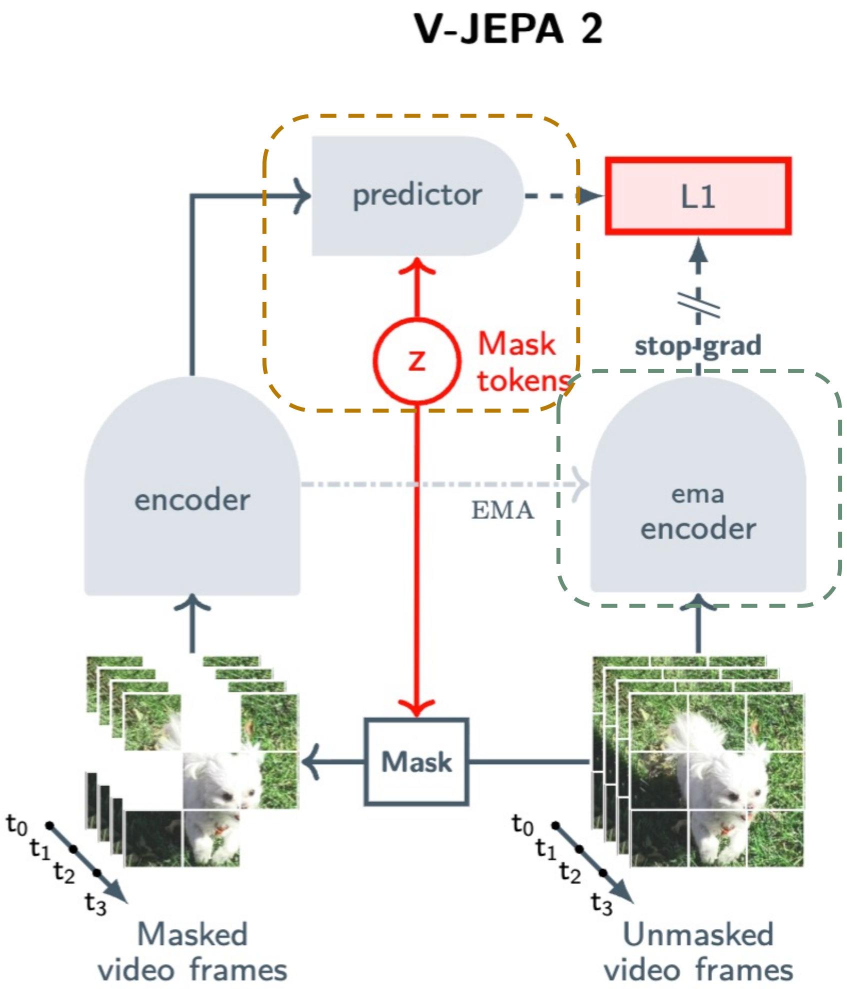

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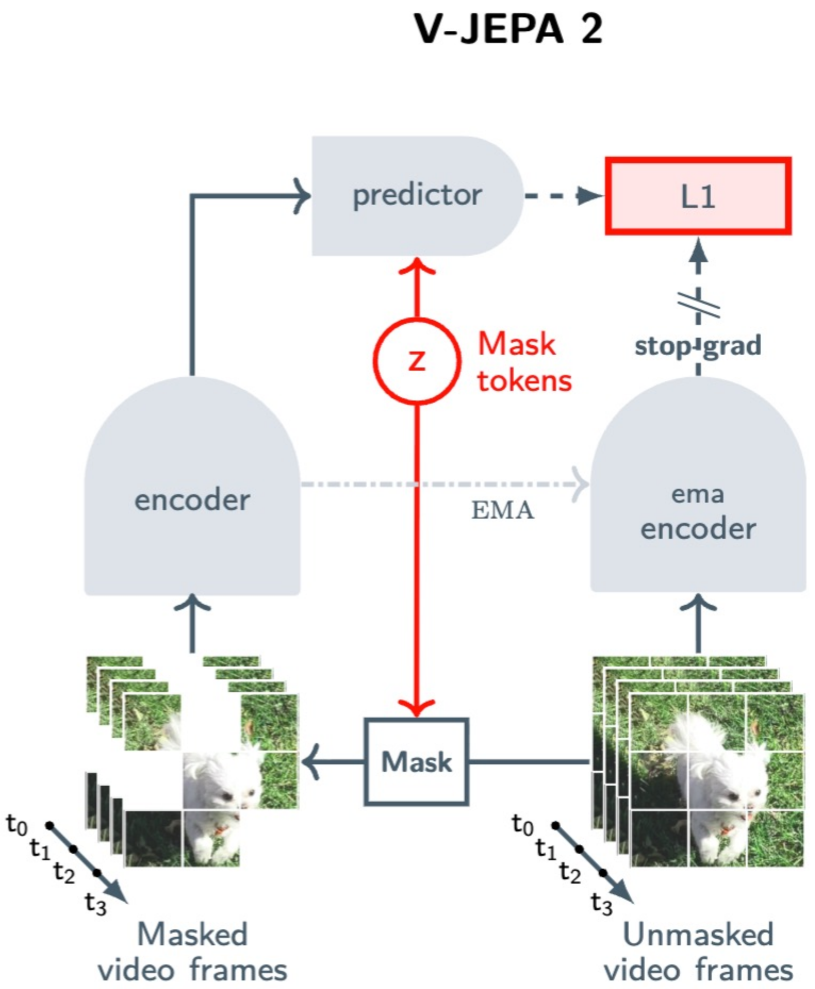
- **Predict missing representations instead of reconstructing pixel**



V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

- Goal: Predict the representation(y) of the original video from a masked view(x)
- Loss:

$$\underset{\theta,\phi,\Delta_y}{\text{minimize}} \left\| P_\phi\left(\Delta_y, E_\theta(x)\right) - \mathrm{sg}\left(E_{\overline{\theta}}(y)\right)\right\|_1$$

Δ_y : Mask tokens (location of dropped patches)
E_θ : Encoder
$E_{\overline{\theta}}$: EMA Encoder
P_ϕ : Predictor
$\mathrm{sg}(\cdot)$: stop gradient operation

- Loss is applied only to **masked patches**

V-JEPA2: Self-Supervised Video Pretraining

Mask-Denoising in Representation Space

- Goal: Predict the representation(y) of the original video from a masked view(x)
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$$\underset{\theta,\phi,\Delta_y}{\text{minimize}} \left\| P_\phi\left(\Delta_y, E_\theta(x)\right) - \text{sg}\left(E_{\bar{\theta}}(y)\right) \right\|_1$$

- Preventing Representation Collapse
  - Stop-gradient blocks gradient flow through the target branch
  - EMA encoder provides stable target representations
  - EMA: $\bar{\theta} = m\bar{\theta} + (1 - m)\theta$

V-JEPA2: Self-Supervised Video Pretraining

Scaling Ingredients

- Data: 2M $\longrightarrow$ 22M videos
- Model: 300M $\longrightarrow$ 1B parameters
- Progressive resolution training

V-JEPA2-AC: Action-Conditioned World Model

Why V-JEPA 2-AC ?

- Limitation of V-JEPA 2 (action-free):
  - Learns visual representations
  - Cannot predict the effect of **actions**
- Goal: Convert V-JEPA 2 into an **action-conditioned latent**
  - Small amount of interaction data
  - Useful for planning and control
- Output:
  - Predict future video representation

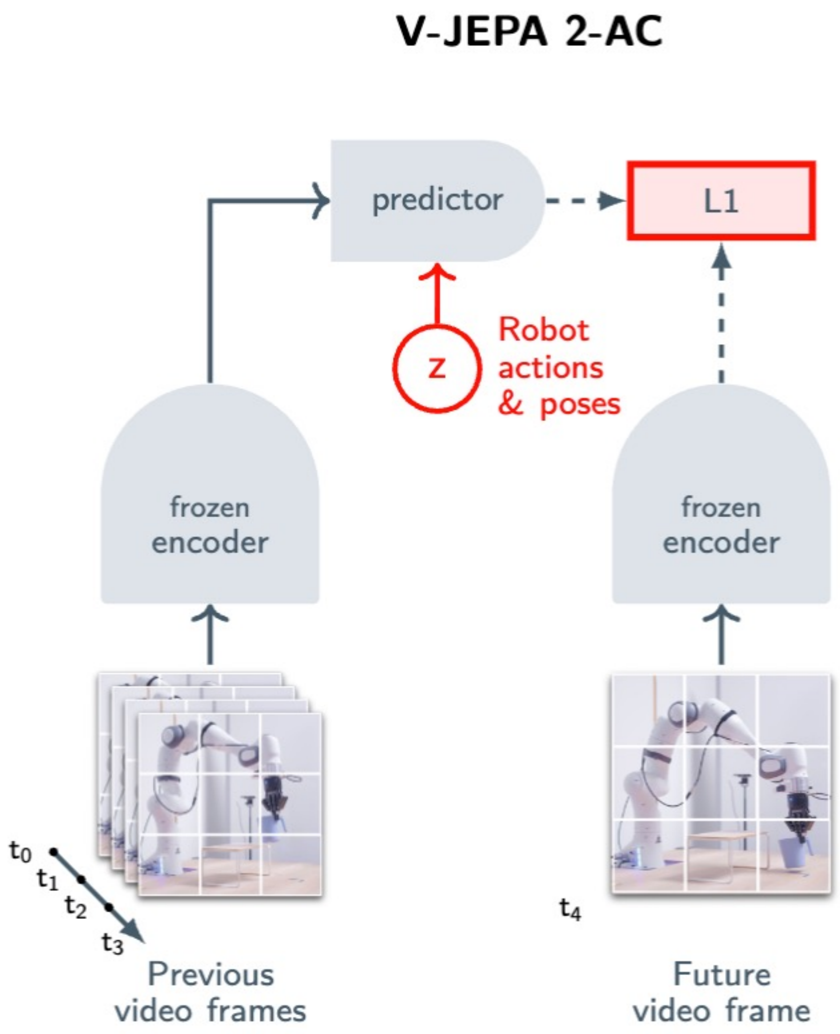
V-JEPA2-AC: Action-Conditioned World Model

Training Data: Droid Dataset

- 62H of unlabeled robot videos
- No reward
- No task labels

Inputs:

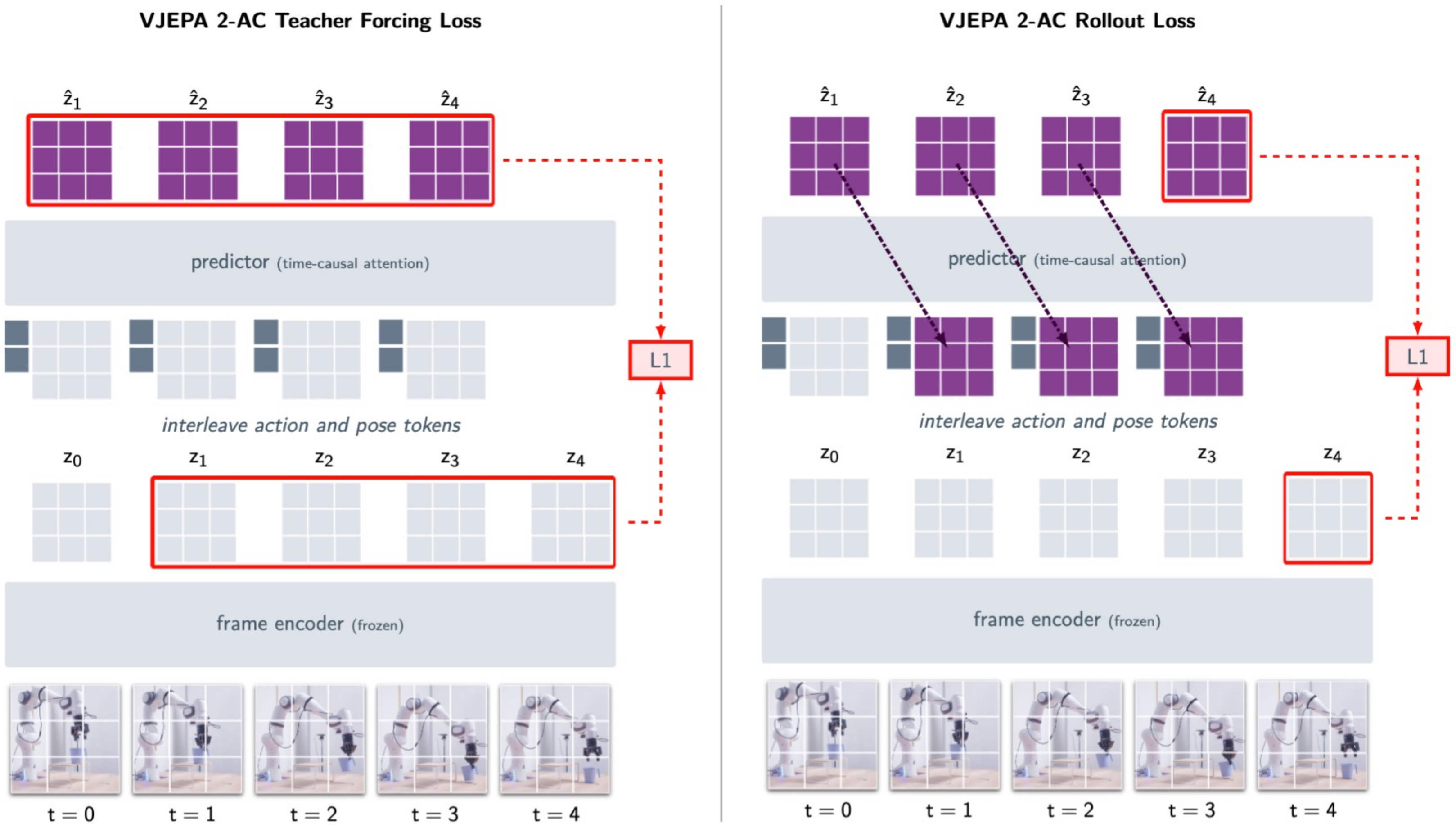
- x_k : Current frame
- s_k : Robot state
  - End-effector state
  - Position (3D) / Orientation (3D) / Gripper state (1D)
- a_k : Action
  - Change in end-effector state



V-JEPA2-AC: Action-Conditioned World Model

Training Objective

1) scalar-valued teacher-forcing loss function
2) Two-step rollout loss



V-JEPA2-AC: Action-Conditioned World Model

Training Objective

1) scalar-valued teacher-forcing loss function

- Goal: predict next representation ($\hat{z}_{k+1}$)

$$\mathcal{L}_{\text{teacher-forcing}}(\phi) := \frac{1}{T}\sum_{k=1}^{T}\|\hat{z}_{k+1} - z_{k+1}\|_1 = \frac{1}{T}\sum_{k=1}^{T}\left\|P_\phi\left(\left(a_t, s_t, E(x_t)\right)_{t\leq k}\right) - E(x_{k+1})\right\|_1$$

- Teacher-Forcing:
  - Input ground-truth state as next input (instead of model prediction)
  - Enables stable learning of one-step dynamics

V-JEPA2-AC: Action-Conditioned World Model

Training Objective

2) Two-step rollout loss

- Goal: improve autoregressive rollout during **inference**

$$\mathcal{L}_{\text{rollout}}(\phi) := \left\| P_\phi(a_{1:T}, s_1, z_1) - z_{T+1} \right\|_{1.}$$

$P_\phi(a_{1:T}, s_1, z_1)$: final predicted state representation

- Motivation:
  - Teacher-forcing uses ground-truth states during training
  - In inference, the model must use its own predictions
  - Prediction errors can accumulate

V-JEPA2-AC: Action-Conditioned World Model

Training Objective

1) scalar-valued teacher-forcing loss function

2) Two-step rollout loss

Overall Training Objective:

$$L(\phi) := \mathcal{L}_{\text{teacher-forcing}}(\phi) + \mathcal{L}_{\text{rollout}}(\phi)$$

- Better long-horizon prediction
- More reliable planning

Planning

From Prediction to Planning

- Training:
  - Current State + Action → Predict Future state
- Planning:
  - Current State + Goal State → Find Action Sequence

Planning

Energy Minimization

- Goal: Find optimize actions that make predicted future state similar to the goal state
- goal-conditioned energy function:

$$\mathcal{E}\left(\hat{a}_{1:T}; z_k, s_k, z_g\right) := \left\| P(\hat{a}_{1:T}; s_k, z_k) - z_g \right\|_1$$

z : feature map (k: current, g: goal)

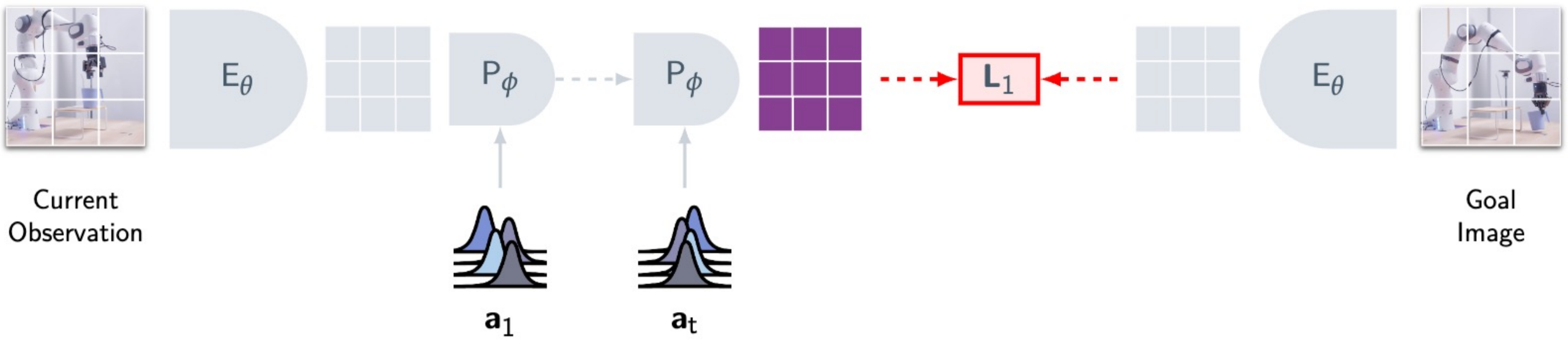
- Action sequence:

$$(a_i^*)_{i \in [T]} := argmin_{\hat{a}_{1:T}} \, \mathcal{E}\left(\hat{a}_{1:T}; z_k, s_k, z_g\right)$$

Planning

Energy Minimization

- In each Planning Step:
    1) Cross-Entropy Method (CEM)
        - Iteratively optimize action sequence through sampling and refinement
    2) Receding Horizon Control
        - Execute only the first action and re-plan using new observations



Results: Planning

Planning Performance & Efficiency

Lab 2	Planning Details					Grasp		Pick-&-Place	
Method	#Samples	Iter.	Horizon	Time	**Reach**	Cup	Box	Cup	Box
Cosmos (Agarwal et al., 2025)	80	10	1	4 min.	80%	0%	20%	0%	0%
V-JEPA 2-AC (ours)	800	10	1	16 sec.	100%	60%	20%	80%	50%

- V-JEPA 2-AC achieves higher success rates across robot manipulation tasks
- Predicting in latent representation:
  - More effective
  - More computationally efficient

Conclusion

- Self-Supervised video pretraining learns strong visual representations
- V-JEPA 2 supports **understanding and prediction** across diverse tasks
- V-JEPA 2-AC enables **planning** and zero-shot robot control
- Demonstrates a unified framework for understanding, prediction and planning from video

Future Works

- Long-horizon Planning
- Language-based Goal Specification
- Scaling